BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

MEDICAL DISCLAIMER: AI-based preliminary estimation only. This result must be confirmed using a standard laboratory blood typing test before medical use. This system does not replace standard serological blood typing.

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000202	**Timestamp:**	2026-09-05 14:13:50
Input Source:	MFS100 Consensus (3 Impressions) | Standard Finger: LEFT THUMB	**Evaluation Status:**	LOW_CONFIDENCE
Model Name:	ResNet50 Transfer Learning	**Confidence:**	**41.61%**
Standard Acquisition Finger:	**LEFT THUMB** (Convention-based protocol — not automatically verified by scanner)		

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



Predicted Blood Group:

O-

Confidence Score: **41.61%**
Threshold: 60.0%
Classification: Low Confidence - Retest Required
Derived ABO Group: **O** (66.2% prob) | Rh Factor: **-**

Rh FACTOR UNCERTAINTY ADVISORY: Rh factor (±) determination is uncertain (narrow Rh probability margin (17.0%); moderate 8-class confidence (41.6%); predicted Rh-negative requires serological validation due to weak ridge-Rh correlation). ABO group appears to be 'O' (66.2% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
0.0%	14.9%	0.0%	18.8%	0.0%	0.0%	24.6%	41.6%